Cost Avoidance Analysis Report

Make or buy assessment | Forecast volume 10,000 units

Recommendation

Continue making in-house under the supplied assumptions. Outsourcing avoids $43,000 of manufacturing costs but adds a $45,000 supplier payment. The $7,000 of manufacturing costs that remain brings the buy option to **$52,000**, compared with **$50,000** to make. Buying therefore increases cost by **$2,000, or $0.20 per unit**.

Scope: one forecast batch or period of 10,000 units. The source does not specify a currency or time horizon; no annualization or currency conversion is applied.

Cost avoidance by category

Manufacturing cost	Make cost	Avoided	Cost avoided	Cost retained
Materials and components	$9,000	100%	$9,000	$0
Direct labor	$12,000	100%	$12,000	$0
Variable factory overhead	$5,000	80%	$4,000	$1,000
Fixed factory overhead	$24,000	75%	$18,000	$6,000
Total	**$50,000**	**86%**	**$43,000**	**$7,000**

Materials and direct labor are treated as fully avoidable, as implied by the source worksheet. The 86% total avoidance rate is $43,000 divided by $50,000.

Comparable total cost

Cost measure	Make in-house	Buy externally
Manufacturing cost incurred	$50,000	$7,000
Supplier purchase payment	$0	$45,000
Total cost for 10,000 units	$50,000	$52,000
Total cost per unit	$5.00	$5.20

Net savings from buying = avoided manufacturing cost - purchase payment
= $43,000 - $45,000 = **-$2,000**.

The worksheet labels this result a $(2,000) variance to purchase. It is an unfavorable buying variance. Gross cost avoidance of $43,000 is not net savings because the replacement purchase payment must also be included.

Decision Thresholds and Assumptions

Break-even analysis | Same 10,000-unit forecast and cost base

Break even purchase price

Maximum supplier price = avoidable manufacturing cost / forecast units
$43,000 / 10,000 = $4.30 per unit. At this price, the $43,000 supplier payment plus $7,000 retained cost equals the $50,000 make cost.

The quoted $4.50 price must fall by **$0.20 per unit, or 4.44%**, to reach parity. A price below $4.30 creates savings before any additional buying costs or capacity benefits.

Supplier price per unit	Total buy cost	Net savings from buying
$4.00	$47,000	+$3,000
$4.30	$50,000	$0 (break even)
$4.50 (proposal)	$52,000	-$2,000
$5.00	$57,000	-$7,000

Sensitivity holds the supplied manufacturing costs and avoidance factors constant. Each $0.10 change in unit purchase price changes the result by $1,000. These thresholds apply to the stated volume.

What could change the decision

Greater fixed overhead avoidance. With variable overhead avoidance held at 80%, fixed overhead savings must reach $20,000: $45,000 - $9,000 - $12,000 - $4,000. This is **83.33%** of $24,000, versus the assumed 75%. At 100% fixed overhead avoidance, buying saves $4,000.

Productive use of released capacity. At the current quote, outsourcing needs at least **$2,000 of incremental net benefit** from released capacity to reach parity. Use net contribution after related costs, rather than sales revenue. Do not count overhead savings twice.

Additional buying costs. Freight, inspection, supplier qualification, transition costs, nonrecoverable duties or taxes, and working capital effects have not been supplied. Include each incremental item before final approval.

Validation before a decision

Confirm that materials and labor spending can actually be eliminated, and that $18,000 of fixed overhead can be removed within the decision period. Reallocating overhead to another product does not create a saving. The $7,000 retained cost continues under buying; it is not necessarily a sunk cost.

Confirm equivalent quality, delivery reliability, supplier capacity, and an all-in price for 10,000 units. The analysis assumes no additional outsourcing costs, no capacity opportunity benefit, and no differences in timing or financing. Any one-time savings or costs must use the same horizon as the operating figures.

Source and method. User-provided image(1).png, Steps 1 to 4. Amounts were transcribed and recalculated. Price sensitivity and break-even thresholds are derived case calculations; the source assumptions have not been independently validated.